



Bhargavi Teja Mudunuru

Sr.Python developer

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Professional Summary

- 8 years of expertise as a software engineer in financial services & product development, known for delivering top-tier work that enhances user experience.
- Implemented & developed **scalable microservices** using **Flask & Django**, optimising API performance with **Gunicorn, Nginx, & Redis caching**.
- Designed & implemented **RESTful & GraphQL APIs**, enabling seamless integration with high-traffic distributed applications.
- Built & deployed **asynchronous task execution pipelines** with **Celery, RabbitMQ, & AWS Step Functions**, ensuring fault-tolerant background processing.
- Developed **database-driven applications** with **PostgreSQL, MySQL, & Redshift**, leveraging **SQL Alchemy ORM & asynchronous query optimisation** using **asyncpg**.
- Implemented **real-time event-driven architectures** using **Apache Kafka, AWS Kinesis, & WebSocket's**, supporting streaming analytics & data-driven applications.
- Engineered **containerised deployments** with **Docker** & orchestrated scalable workloads using **Kubernetes (EKS), Helm charts, & Kustomize**.
- Integrated **CI/CD pipelines** with **GitHub Actions, Jenkins, & AWS Code Pipeline**, automating build, test, & deployment processes for production-ready applications.
- Secured APIs & backend systems with **OAuth2, JWT, AWS Cognito, & IAM policies**, ensuring compliance with enterprise security standards.
- Designed **batch & real-time data processing workflows** using **Pandas, NumPy, PySpark, & AWS Glue**, improving ETL efficiency for high-volume datasets.
- Developed & optimised **Django ORM queries** using **prefetch-related, select-related, & raw SQL**, reducing query execution time by **40%**.
- Integrated **AWS Lambda & API Gateway** for serverless computing, reducing infrastructure costs while maintaining application performance.
- Implemented **observability & monitoring** with **ELK Stack (Elasticsearch, Logstash, Kibana), AWS CloudWatch, & Open Telemetry**, enabling real-time system diagnostics.
- Designed **role-based access control (RBAC) models**, securing sensitive data across multiple environments using **AWS Lake Formation & IAM**.
- Led application refactoring initiatives by **migrating monolithic Django applications to microservices**, improving scalability, maintainability, & deployment efficiency.
- Developed **full-stack solutions** with **React, Django REST Framework (DRF), & Flask**, enhancing user experience & backend performance.
- Automated test-driven development (TDD) workflows with **Pytest, Unit test, & CI/CD integrations**, improving software reliability & deployment stability.
- Designed **data validation & schema evolution mechanisms** using **Pydantic, Marshmallow, & Apache Iceberg**, ensuring seamless data integrity in production pipelines.
- Configured **gRPC-based microservices** for optimised inter-service communication, reducing API latency & enhancing system scalability.

Technical Skills

Programming Languages	Python 2.X, Python 3.X, Pyspark, C, C++, Java, J2EE, SQL, R, UNIX.
Web Technologies	HTML, CSS, XHTML, Java Script, jQuery, AJAX, XML, JSON, Node JS.
Webserver	WebSphere, Web logic, Apache Tomcat, Oracle Web Logic.
Databases	AWS Dynamo DB, MySQL, PL/SQL, Oracle, Microsoft SQL, PostgreSQL, MongoDB, MariaDB, NoSQL, Pig, Hive, Map Reduce.
Python Web Frameworks	Django, Flask, Fast API, Spring MVC, Hibernate, Spring Boot, Kafka, Kubernetes,

	Docker, Scala, Maven, Kubernetes.
Development IDE	PyCharm, Eclipse, Net beans, MS Visio, Notepad++.
Operating Systems	Linux, Windows.
Cloud & Web Services	SOAP, RESTful, AWS, GCP.
Version Control	Git-Hub.
Build Tools	Apache Maven, Apache Ant, Apache Spark, GNU, WinSCP, Putty, TFS, GIT, Jenkins, Docker.
Methodologies	Agile, Scrum, Waterfall.

Experience

BCBS, Chantilly, Virginia (Remote)

July '24 – Till Date

Python Developer

Responsibilities:

- Designed & implemented **microservices-based architecture** using **Django REST Framework (DRF)**, enabling independent scalability & deployment.
- Developed **GraphQL APIs** with **Django** & integrated **Apollo Client** on the front end, improving query efficiency & reducing data over fetching.
- Optimised **PostgreSQL** database performance by implementing **indexing, partitioning, & materialised views** for handling large-scale transactional data.
- Integrated **AWS Cognito** for authentication & authorisation, implementing fine-grained access controls using **JWT & OAuth2**.
- Containerised the application using **Docker** & deployed it on **AWS ECS**, ensuring seamless horizontal scaling with **Fargate**.
- Implemented background task processing with **Celery & Redis**, optimising performance for asynchronous operations like report generation & email notifications.
- Leveraged **AWS S3** for object storage, integrating **pre-signed URLs** for secure & efficient data retrieval.
- Configured & optimised **CloudFront** as a **CDN** to enhance static asset delivery & improve response times globally.
- Designed a multi-region data replication strategy using **AWS DMS (Database Migration Service)** for high availability & disaster recovery.
- Implemented real-time data monitoring & alerting using **AWS CloudWatch, Prometheus, & Grafana**, ensuring proactive performance tuning & anomaly detection.
- Developed an end-to-end **CI/CD pipeline** using **GitHub Actions & AWS Code Pipeline**, ensuring automated testing, security scans, & zero-downtime deployments.
- Monitored & optimised application performance using **AWS X-Ray** for tracing **API requests** & identifying latency bottlenecks.

Environment: Python, Amazon AWS S3, MySQL, HTML, PySpark, Python 2.7, Django 1.4, HTML5, CSS, XML, PyCharm, Notebook, MySQL, MS SQL Server, JavaScript, AWS, Linux, Shell Scripting, AJAX.

Development Bank of Singapore (DBS), India

July '21 – January '23

Senior Software Developer

Responsibilities:

- Conducted requirement gathering & analysis with business stakeholders, documenting project requirements.
- Designed & implemented scalable **data ingestion pipelines** using **Python, Apache Kafka, & AWS Kinesis** to handle high-throughput streaming data.
- Developed **Flask-based API microservices** for real-time data processing, integrating with **AWS Lambda** for event-driven execution.
- Implemented advanced **ETL pipelines** using **Apache Airflow**, orchestrating data transformations & loading structured/unstructured data into **AWS Redshift & S3**.
- Optimised **Spark jobs** on **AWS EMR** by fine-tuning **memory allocation, shuffle operations, & dynamic partition pruning** for efficient big data processing.

- Leveraged **AWS Glue Data Catalog** to maintain **metadata & schema evolution** across various data sources.
- Designed partitioned & **columnar storage models in Parquet format**, improving read performance in **Redshift Spectrum & Athena**.
- Implemented distributed message processing using **Kafka Streams**, enabling real-time event-driven data transformations & efficient high-velocity data processing.
- Developed schema evolution strategies with **Apache Avro & Confluent Schema Registry**, ensuring backward & forward compatibility in streaming pipelines.
- Optimised **Lambda functions** by leveraging Provisioned Concurrency, reducing cold start latency for real-time API & event processing.
- Implemented automated **data validation & anomaly detection** & integrated **AWS CloudWatch** for real-time alerting.
- Developed **real-time analytics dashboards** by streaming pre-aggregated data to **Amazon OpenSearch Service** (formerly **Elasticsearch**).
- Optimised **data workflows** using **Step Functions & AWS Batch**, enabling parallel execution of heavy compute jobs with **Spot Instances** to reduce costs.
- Implemented fine-grained **data access policies** using **AWS Lake Formation & IAM policies**, ensuring role-based permissions for sensitive datasets.

Environment: Python, Amazon AWS S3, Lambda, Dynamo DB, Java, J2EE, Unix, Docker, Kubernetes, MariaDB, MySQL, PySpark, Python 2.7, HTML5, CSS, XML, MySQL, Kafka, PyCharm, Jupiter Notebook.

Achievements:

- Received appreciation from the Business users & Project Managers for the timely completion of tasks.
- Won “TOP 20 ENGINEERS of the Bank” multiple times.
- Received “Star Performer” award.

Cognizant Technology Solutions, India
Software Developer

Jan '17 – July '21

Responsibilities:

- Implemented **scalable microservices** using **Flask & Django**, implementing **modular architecture & RESTful APIs** for seamless service communication.
- Built **asynchronous task execution** using **Celery with Redis & RabbitMQ**, ensuring **fault-tolerant background processing** for data-intensive workflows.
- Optimised **Flask API performance** by leveraging **Gunicorn workers, caching strategies (Redis, Memcached), & Nginx as a reverse proxy**.
- Integrated **SQLAlchemy ORM with PostgreSQL & MySQL**, utilising **asynchronous database connections** via **asyncpg** for optimised query execution.
- Implemented **API rate limiting, authentication, & authorisation** using **OAuth2, JWT, & AWS Cognito**, ensuring secure client access & role-based permissions.
- Developed **GraphQL APIs with Flask & Django** to enable **flexible, efficient data querying** while reducing payload size & request overhead.
- Built **full-stack applications** with **React & Django**, integrating **Django REST Framework (DRF)** for backend API services.
- Designed & executed **unit & integration tests** using **Pytest & Unit test**, achieving high test coverage & minimising regression risks.
- Leveraged **WebSocket's with Flask-Socket IO** to enable **real-time notifications, chat functionality, & live dashboards** for streaming data.
- Deployed containerised applications using **Docker** & orchestrated workloads with **Kubernetes (EKS)**, optimising resource allocation with **Helm charts & Kustomize**.
- Integrated **CI/CD pipelines** using **GitHub Actions & AWS Code Pipeline**, automating build, test, & deployment processes for Flask & Django applications.
- Engineered **efficient SQL queries & stored procedures** for **PostgreSQL, MySQL, & Redshift**, optimising database performance for large-scale datasets.

- Refactored **monolithic Django applications into microservices**, improving maintainability & scalability & reducing deployment downtime.
- Implemented **asynchronous task execution** using **Django Channels**, enabling **real-time event-driven architectures** for high-performance applications.
- Optimised **Django ORM queries** using **prefetch related, select related, & raw SQL queries**, reducing API response times by 40%.
- Designed **real-time streaming data pipelines** using **Apache Kafka & AWS Kinesis**, integrating with **Flask-based services** for live analytics.
- Implemented **structured logging & monitoring systems** using **ELK Stack (Elasticsearch, Logstash, Kibana) & AWS CloudWatch**, improving observability & debugging.
- Configured **Swagger/Open API documentation** for **Flask & Django APIs**, enabling seamless integration with third-party clients & internal teams.
- Developed **batch data processing applications** using **Pandas & NumPy**, integrating with **AWS Lambda & Step Functions** for automated ETL workflows.
- Designed **GraphQL & gRPC-based microservices**, reducing API latency & enabling high-performance, structured communication across distributed systems.

Environment: Python, Java, J2EE, Unix, Docker, MySQL, PySpark, Python 2.7, Django, Flask, Pandas, NumPy, SQL Alchemy, Unit test, Pytest, Gunicorn, Jingo 2, Web Services, Hibernate, Spring, HTML, SQL, UNIX, CSS, XML, MySQL, SQL Server 2012/2008 R2, packages, Visual studio 2010/2008, Agile, Docker, Kubernetes.

ITC Infotech Hyderabad, India

May '16 – Dec '16

Python Developer

Responsibilities:

- Designed the front end of the application using **Python, Java, HTML, CSS, AJAX, JSON and jQuery**. Worked on backend of the application, mainly using Active Records.
- Involved in the design, development and testing phases of application using **AGILE methodology**.
- Maintained databases using **Python** and developed **Python based API (RESTful Web Service)** using **Flask, SQL Alchemy and PostgreSQL**.
- Worked on development of **SQL** and **stored procedures, trigger and function on MYSQL**.
- Developed **server-based web traffic** statistical analysis tool using **Flask, Pandas**.
- Used **Python and Django** creating **graphics, XML processing of documents, data exchange and business logic** implementation between servers.
- Created restful web services for Catalog and Pricing with **Django MVT, Jersey, MySQL, and MongoDB**.
- Used **MySQL** as backend database and **MySQL dB of python** as database connector to interact with **MySQL** server.
- Used **Numpy** for **Numerical analysis, PySpark to expose Spark API** to **Python**.
- Developed applications using **MongoDB, Redis, and MySQL**.
- Involved in writing application level code to interact with **APIs, Web Services** using **JSON**.
- Involved in **Agile Methodologies and SCRUM** Process.
- Developed entire frontend and backend modules using **Python on Django Web Framework**.
- Actively involved in developing the methods for **Create, Read, Update and Delete (CRUD)** in **Active Record**.
- Used **MVC Architecture** for developing the application.

Environment: Python, Java, MySQL, Spring MVC, Django, API, HTML, CSS.

Achievements:

- Received appreciation from the client & Project Managers for the timely completion of tasks.
- Northern Trust was awarded a "Wall of Fame" for its development activities.
- Received "Star Performer" awards thrice in a row at Cognizant.
- Received the "Best Team Player" award multiple times for working on the most issues.

Education

The University of Texas at Dallas

Master of Science, Information Technology & Management

January '23 – May '24

GPA: 3.85/4.0

Jawaharlal Nehru Technological University Kakinada, India

Bachelor of Technology, Electronics & Communication Engineering

September '12 – May '16

GPA: 4.0/4.0

Certifications

- AWS Certified Solutions Architect
- AWS Certified Cloud Practitioner
- AWS Certified Developer Associate
- Google Data Analytics Certification
- Oracle Cloud Infrastructure 2023 Certified Foundations Associate
- Oracle Cloud Data Management 2023 Certified Foundations Associate
- Banking & financial services Domain certification